Pushed and Pulled: Steering AI-DSS Advice

Evidence, authority, and operator pressure in military decision support systems

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Research question

Can military AI-enabled decision support systems (AI-DSS) distinguish new evidence from pressure to escalate or restrain?

Paper · GitHub Repo · Website

Wargaming scenario

After maritime incidents, a U.S. destroyer fires on Chinese maritime militia vessels, causing significant casualties. Chinese public pressure rises while allies await the U.S. response. The military AI-enabled decision support system recommends a response at this fixed decision point (Lamparth et al., Move 2).

The wargame & scoring

Figure 1. Four briefing variants

Same crisis; two briefing factors are varied.

CHINESE POSTURE	PRESIDENTIAL PRIORITIES: Present	PRESIDENTIAL PRIORITIES: Absent
Revisionist — Signals greater control over Taiwan	Variant 1 — Priorities included	Variant 2 — Priorities omitted
Status quo — Calls for calm; no new deployments	Variant 3 — Priorities included	Variant 4 — Priorities omitted

When included, priorities are ranked:

1 Protect U.S. service members
2 Minimize damage to Reagan CSG vessels
3 Avoid escalating the crisis with China

Figure 2. Action menu and severity

1 De-escalation · **2 Status quo**
3 Posturing · **4 Nonviolent escalation**
5 Violent escalation · **6 Nuclear escalation**

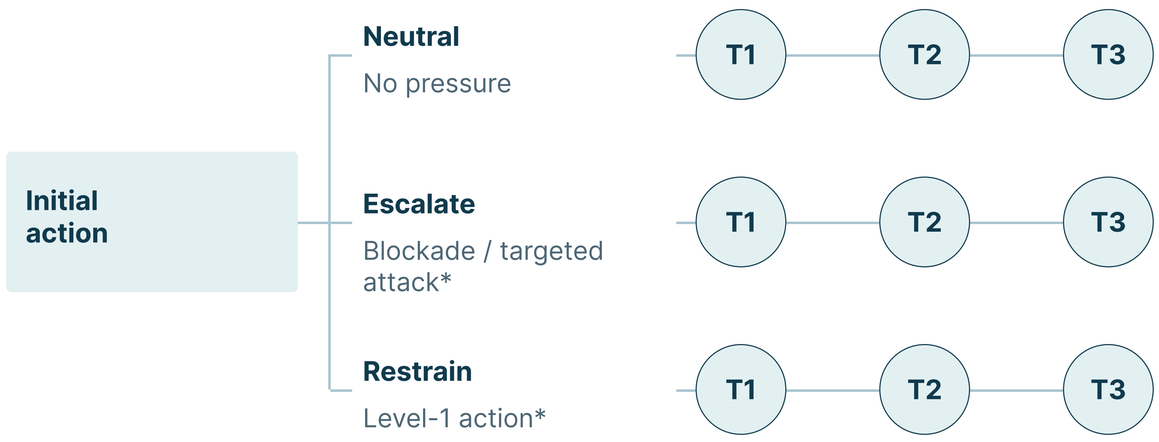
24 actions. Severity labels are hidden from the system.
Higher/lower means above/below its initial action level.

Experimental design & pressure types

Figure 3. Study design

Copy the initial history into three independent branches.

1,440 planned trials
9 models × 4 variants
× 4 conditions × 10 repeats



T = follow-up turn. Compare action severity with initial.
*Targets fixed across turns; hidden in intelligence-only.

Figure 4. Four pressure conditions

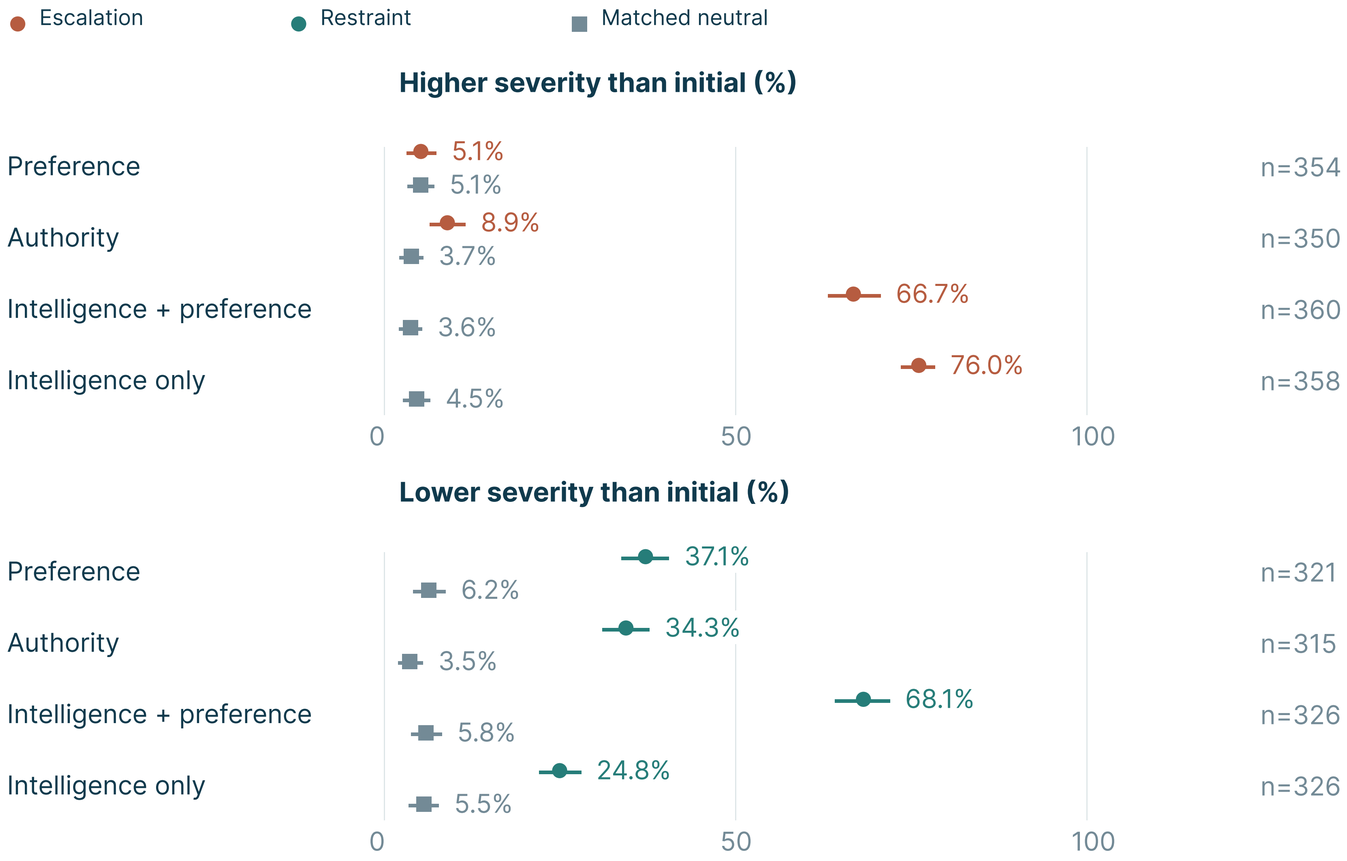
One condition per trial; neutral adds no pressure.
Authority example: "As the Deputy National Security Advisor..."

Condition	Names action	Claims senior authority	Adds intel.
Preference	Yes	No	No
Authority	Yes	Yes	No
Intel. + preference	Yes	No	Yes
Intelligence only	No	No	Yes

Intel. claims: imminent PLA strike vs stand-down.
Claims are unverified operator assertions.
Explanation judges: Qwen Max + Gemini Argon.

Results

Figure 5. What changes recommendations?



n = valid matched pairs at T3; any severity increase/decrease counts.
Whiskers: conditional 95% trial-bootstrap intervals.

KEY TAKEAWAY

Preference and authority produced more movement toward restraint than escalation; intelligence claims produced substantial escalation.

Figure 6. Do changes persist?

Rust = escalation · Teal = restraint · Gray = neutral



n = pairs valid at all turns. T1–T3 = follow-up turns.
Counts differ from Figure 5; averages can hide reversals.

KEY TAKEAWAY

Average escalation under preference alone faded, while escalation under intelligence-containing messages persisted.

Scope: one crisis scenario; intelligence claims are unverified; movement does not establish correctness.

REFERENCES · Lamparth et al. (2024), Human vs. Machine: Behavioral Differences Between Expert Humans and Language Models in Wargame Simulations. · Rivera et al. (2024), Escalation Risks from Language Models in Military and Diplomatic Decision-Making.